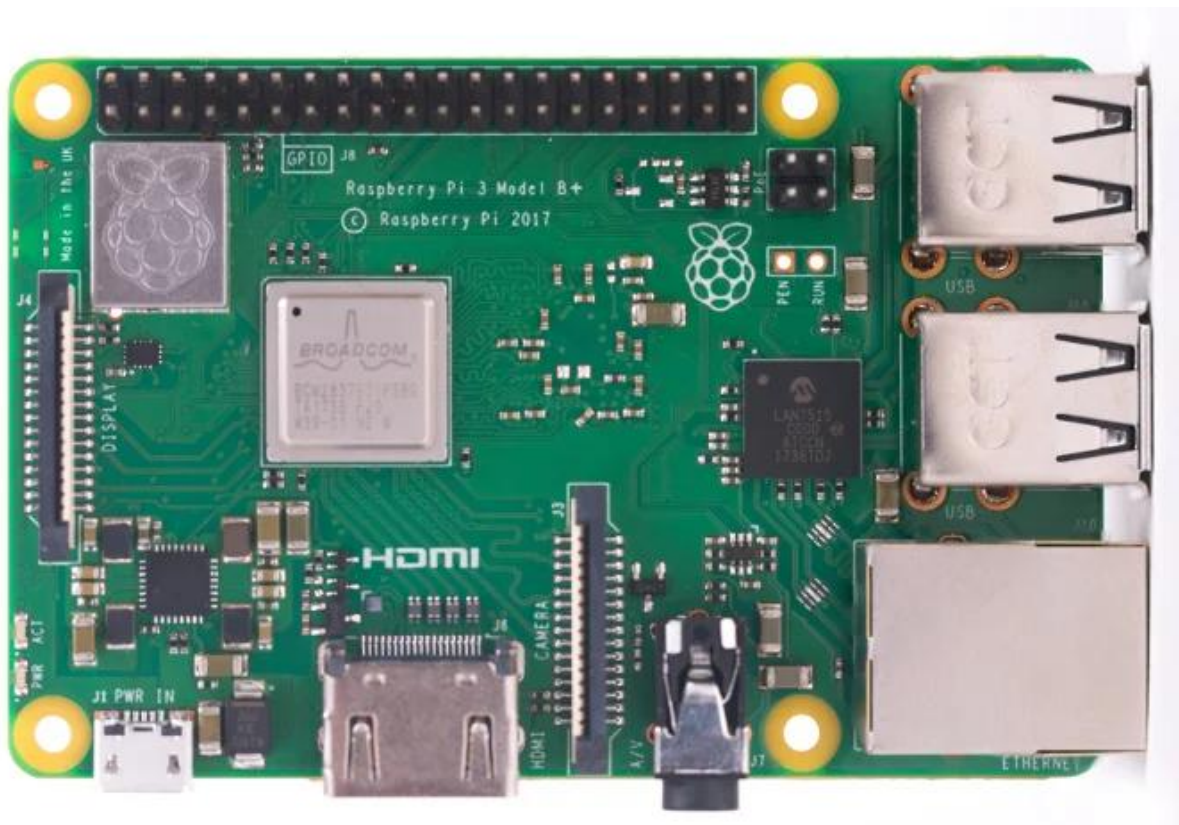


Internet Of Things

RASPBERRY PI

Overview

Pi is a single-board computer". Pi is a small scale computer in the size little bigger than a credit card, it packs enough power to run games, word processor like open office, image editor like Gimp and any program of similar magnitude. Pi was introduced as an educational gadget to be used for prototyping by hobbyists and for those who want to learn more about programming. It certainly cannot be a substitute for our day to day Linux, Mac or Windows PC. Pi is based on a Broadcom SoC (System of Chip) with an ARM processor [~700 MHz], a GPU and 256 to 512 MB RAM. The boot media is an SD card [which is not included], and the SD card can also be used for persist data. Now that you know that the RAM and processing power are not nearly close to the power house machines you might have at home, these Pi's can be used as a Cheap computer for some basic functions, especially for experiments and education. The Pi comes in three Configurations and we will discuss the specifications of those in the coming sections. The cost of a Pi is around \$35 for a B Model and is available through many online and physical stores.



Raspberry pi board

System specifications

Computer	A Raspberry Pi
Storage	SD Card and a SD card reader to image the OS [These days laptops have inbuilt card readers]
Power supply	5 volt micro USB adapter, mostly your android phone charger would work

Display	An TV/Monitor with DVI or HDMI port
Display connector	HDMI cable or HDMI to DVI converter cable
Input	USB Mouse
Input	USB Keyboard
Network	Ethernet cable
Case	If you really need one, you can get them online based on the model you have

Raspberry Pi uses

It can be used in following areas/applications:

1. Retro Gaming
2. Raspberry Pi Tablet
3. Low-Cost Desktop PC
4. Raspberry Pi Cluster
5. Raspberry Pi Cloud Server
6. Raspberry Pi Media Center
7. Web Server
8. Home Automation System
9. VPN
10. Robotics

MCQ QUESTIONS

1. Does Raspberry Pi need external hardware?
 - a) True
 - b) False
2. Does RPi have an internal memory?
 - a) True
 - b) False
3. What do we use to connect TV to RPi?
 - a) Male HDMI
 - b) Female HDMI
 - c) Male HDMI and Adapter
 - d) Female HDMI and Adapter
4. How power supply is done to RPi?
 - a) USB connection
 - b) Internal battery
 - c) Charger
 - d) Adapter
5. What is the Ethernet/LAN cable used in RPi?
 - a) Cat5
 - b) Cat5e
 - c) Cat6
 - d) RJ45
6. What are the parameters that are default values?
 - a) Port_Name and Bits
 - b) Speed and Port_Names
 - c) Speed and Parity
 - d) Stop bit and Flow Control
7. What is the command used for easy using of GNU screen?
 - a) `$useradd -G {dialout} your_name`
 - b) `Screen Port_Name115200`
 - c) `Minicom -b 115200 -o -D Port_Name`
 - d) `Prompt> # help`
8. Which instruction set architecture is used in Raspberry Pi?
 - a) X86
 - b) MSP
 - c) AVR
 - d) ARM

9. What is the default user in Debain on Raspberry Pi?

- a) Default
- b) User
- c) Pi
- d) Root

10. What are the distributions are supported by raspberry Pi?

- a) Arch Linux
- b) Debain
- c) Fedora Remix
- d) Arch Linux, Debain, and Fedora Remix

11. What bit processor is used in Pi 3?

- a) 64-bit
- b) 32-bit
- c) 128-bit
- d) Both 64 and 32 bit

12. What is the speed of operation in Pi 3?

- a) 900MHz
- b) 1.2GHz
- c) 1GHz
- d) 500MHz

13. WiFi is not present in which of the following models?

- a) Raspberry Pi3
- b) Raspberry Pi Zero WH
- c) Raspberry Pi Zero W
- d) Raspberry Pi Zero

14. Does micro SD card present in all modules?

- a) True
- b) False

15. How many USB ports are present in Raspberry Pi 3?

- a) 5
- b) 2
- c) 4
- d) 3

Binary conversions in logic system design

Links provided are

<https://www.youtube.com/watch?v=Qnv9vw379dQ>